Prashant

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Education

Vellore Institute of Technology Bhopal, India
Bachelor of Technology - Computer Science and Engineering; CGPA: 8.72 *Sep 2023 - May 2027*
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Databases, Computer Networks

Skills

- **Programming Languages:** C++ , Python , JavaScript (ES6+) , SQL , Object-Oriented Programming (OOP) , Concurrency
- **Python Libraries & ML:** NumPy , Pandas , Matplotlib , Scikit-learn , Google Gemini API , Data Analysis
- **Backend Engineering:** Node.js , Express.js , RESTful APIs , Redis (Caching & Rate-Limiting) , JWT Auth , Nodemailer
- **Databases & ORM:** PostgreSQL , Prisma ORM , MongoDB , Mongoose , Redis , Schema Design , ACID Transactions
- **Core CS Concepts:** Data Structures, Algorithms, Graph Theory, DBMS, Operating Systems, Computer Networks
- **DevOps & Developer Tools:** Git, GitHub, Postman, Docker, Vercel, Render, Google Cloud Console, Neon DB

Projects

- **sPLIT — AI-Powered Expense Splitting & Settlement Engine GitHub | Live Demo**
  - Architected a full-stack expense sharing platform using React, Node.js/Express, Prisma ORM, and Neon Serverless PostgreSQL.
  - Engineered an optimal debt-simplification engine implementing a greedy min-max matching algorithm, collapsing N-way circular debts from $O(N^2)$ to minimum transactions in $O(N \log N)$ time.
  - Integrated Google Gemini 1.5 Flash Multimodal Vision API to parse receipt and bill images, extracting merchants, itemized prices, and tax breakdown into structured JSON schemas.
  - Built 1-click mobile UPI intent routing (`tez://`, `phonepe://`) with dynamic desktop QR generation for zero-friction peer-to-peer debt settlements.
  - Implemented the Cache-Aside pattern via Redis/in-memory cache for group net-balance calculations, bypassing repeated SQL aggregations and cutting read latency from $\sim$65ms to $<$ 4ms.
  - Configured Google OAuth 2.0 with cryptographic ID token validation alongside transactional welcome and settlement emails via Resend API.

  Tech: React, Node.js, Express.js, PostgreSQL, Prisma, Redis, Gemini Vision API, Google OAuth, Tailwind CSS, REST APIs
- **FinLedger Pro — Backend Banking Infrastructure GitHub | Live Demo**
  - Architected a financial ledger microservice implementing strict double-entry accounting with atomic `CREDIT/DEBIT` pairs and Mongoose immutability hooks to prevent ledger tampering.
  - Guaranteed ACID compliance across multi-document transfers via MongoDB sessions (`startSession`) with automatic rollbacks, enforcing UUID-based idempotency keys to eliminate double-spending.
  - Integrated Redis via the Cache-Aside pattern to cache dynamic balance calculations, bypassing expensive database aggregations and slashing read latency from $\sim$45ms to $<$ 1ms.
  - Engineered custom distributed rate-limiting middleware in Redis to protect authentication and transaction endpoints against brute-force attacks and abuse.
  - Implemented secure JWT authentication with TTL-indexed token blacklisting for instant session revocation, paired with asynchronous email notifications via Nodemailer.

  Tech: Node.js, Express.js, MongoDB, Mongoose, Redis, JWT, Nodemailer, RESTful API Design, System Architecture

Certifications

- **Google IT Support Professional Certificate** – Google (Google Skills) Jan 2026
- **The Bits and Bytes of Computer Networking** – Google (Coursera) Dec 2025
- **AWS Technical Essentials** – AWS Educate (Amazon) June 2025
- **Operating Systems and Systems Programming** – Cursa Apr 2025
- **Frontend Development Libraries** – freeCodeCamp Sep 2024
- **Introduction to Machine Learning** – NPTEL (IIT) Sep 2025
- **Market Analysis** – NPTEL (IIT) May 2026

Positions of Responsibility

Cultural Representative – Pahadi Club, VIT Bhopal, India April 2025 – Present
- Led cultural events, fostering teamwork, collaboration, and cross-functional coordination across campus initiatives.